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Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Question Paper Name :

Paper II Applied Science and Engineering
and Management

Subject Name :

Paper II Applied Science and Engineering
and Management

Actual Answer Key :

Yes

Calculator :

None

Magnifying Glass Required? :

No

Ruler Required? :

No

Eraser Required? :

No

Scratch Pad Required? :

No

Rough Sketch/Notepad Required? :

No

Protractor Required? :

No

Show Watermark on Console? :

Yes

Highlighter :

No

Auto Save on Console?

Yes

Change Font Color :

No

Change Background Color :

No

Change Theme :

No

Help Button :

No

Show Reports :

No

Show Progress Bar :

No

Is this Group for Examiner? :

No

Examiner permission :

Cant View

Show Progress Bar? :

No

Paper II Applied Science and Engineering and Management

Section type :

Online

Section Negative Marks :

0.33

Enable Mark as Answered Mark for Review and Clear Response :

Yes

Maximum Instruction Time :

0

Is Section Default? :

null

Question Number : 1 Question Id : 630680395717 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which is the correct order of filling of orbitals according to Aufbau principle?

Options :

1. ✖ $6p < 5d < 7s < 5f$

2. ✔ $5d < 6p < 7s < 5f$

3. ✖ $5d < 7s < 6p < 5f$

4. ✖ $6p < 7s < 5d < 5f$

Question Number : 2 Question Id : 630680395718 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The maximum number of electrons can be accommodated in 'L' shell are:

Options :

1. ✔ 8

2. ✖ 2

3. ✖ 4

4. ✖ 16

**Question Number : 3 Question Id : 630680395719 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

Statement-I: Bohr's model of the hydrogen atom was better than the Rutherford's nuclear model.
Statement-II: Bohr's model could explain the ability of atoms to form molecules by chemical bonds.

Options :

1. ✖ Both the Statements are true.

2. ✖ Both the Statements are false.

3. ✔ Statement I is true but Statement II is false.

4. ✖ Statement I is false but Statement II is true.

**Question Number : 4 Question Id : 630680395720 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

What will be the molarity of NaOH in the solution prepared by dissolving its 16 g in enough water to form 500 ml of the solution?

Options :

1. ✖ 0.4 M

2. ✔ 0.8 M

3. ✖ 0.2 M

4. ✖ 0.1 M

**Question Number : 5 Question Id : 630680395721 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

What is the mass percentage of carbon, hydrogen and oxygen in Phenol?

Options :

1. ✖ 76.59%, 17.02% and 6.38%, respectively
2. ✔ 76.59%, 6.38% and 17.02%, respectively
3. ✖ 76.59%, 3.19% and 20.21%, respectively
4. ✖ 76.59%, 5.38% and 18.02%, respectively

**Question Number : 6 Question Id : 630680395722 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

List-I (Theory)	List-II (Base)
(A) Arrhenius theory	(I) NaOH
(B) Bronsted – Lowry theory	(II) BF ₃
(C) Lewis acid base theory	(III) NH ₃

Choose the correct answer from the options given below:

Options :

1. ✖ A-(I), B-(II), C-(III)
2. ✔ A-(I), B-(III), C-(II)
3. ✖ A-(II), B-(I), C-(III)
4. ✖ A-(II), B-(III), C-(I)

**Question Number : 7 Question Id : 630680395723 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

What will be the molar solubility of $\text{Cu}(\text{OH})_2$ in 0.2 M NaOH. The ionic product of $\text{Cu}(\text{OH})_2$ is 1.6×10^{-19} .

Options :

1. ✓ 4.0×10^{-18} M

2. ✗ 4.0×10^{-19} M

3. ✗ 2.0×10^{-18} M

4. ✗ 2.0×10^{-20} M

Question Number : 8 Question Id : 630680395724 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following statement is NOT true about buffer solutions?

Options :

1. ✗ The solutions which resist change in pH on dilution or with the addition of small amounts of acid or alkali are called Buffer solutions.

2. ✗ The examples of Buffer solutions are Blood and urine.

3. ✗ The mixture of acetic acid and sodium acetate acts as buffer solution around pH 4.75.

4. ✓ The mixture of ammonium chloride and ammonium sulphate acts as a buffer around pH 9.25.

Question Number : 9 Question Id : 630680395725 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (Species)	List-II (Bond order)
(A) O ₂	(I) 2
(B) N ₂	(II) 1
(C) F ₂	(III) 3

Choose the correct answer from the options given below:

Options :

1. ✖ A-(I), B- (II), C-(III)

2. ✔ A-(I), B-(III), C-(II)

3. ✖ A-(II), B-(I), C-(III)

4. ✖ A-(II), B-(III), C-(I)

Question Number : 10 Question Id : 630680395726 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The concentration of the ore 'Copper glance' will be carried out by _____.

Options :

1. ✖ Magnetic Separation

2. ✔ Froth Floatation Method

3. ✖ Hydraulic Washing

4. ✖ Leaching

Question Number : 11 Question Id : 630680395727 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (Ore)	List-II (Metal)
(A) Siderite	(I) Copper
(B) Malachite	(II) Zinc
(C) Calamine	(III) Iron

Choose the correct answer from the options given below:

Options :

- ✖ A-(I), B- (II), C-(III)
- ✖ A-(I), B-(III), C-(II)
- ✔ A-(III), B-(I), C-(II)
- ✖ A-(III), B-(II), C-(I)

Question Number : 12 Question Id : 630680395728 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Statement-A: van Arkel method is very useful for removing all the oxygen and nitrogen present in the form of impurity in certain metals like Zr and Ti.

Statement-R: In van Arkel method, the metal is converted into its iodide as it is more covalent and volatilizes and then decomposed it on a tungsten filament.

Options :

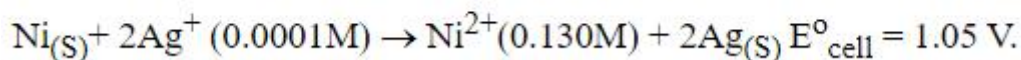
- ✔ Both the statements are true.
- ✖ Both the statements are false.
- ✖ Statement A is true but Statement R is false.
- ✖ Statement A is false but Statement R is true.

Question Number : 13 Question Id : 630680395729 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Calculate the E_{cell} of the cell in which the following reaction takes place.



Options :

1. ✖ 1.68 V

2. ✔ 0.84 V

3. ✖ 1.26 V

4. ✖ 2.96 V

Question Number : 14 Question Id : 630680395730 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The correct increasing order of the oxidizing metals in the electrochemical series is:

Options :

1. ✔ Ag < Cu < Zn < Mg

2. ✖ Cu < Ag < Mg < Zn

3. ✖ Mg < Zn < Ag < Cu

4. ✖ Mg < Zn < Cu < Ag

Question Number : 15 Question Id : 630680395731 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The conductivity of electrolytic(ionic) solutions depends on:

- I. The nature of the electrolyte added
- II. Size of the ions produced and their salvation
- III. The nature of the solvent and its viscosity
- IV. Concentration of the electrolyte
- V. Temperature

The correct statement/s from the above is/are:

Options :

1. ✖ I, II, III only
2. ✖ II, III, IV and V only
3. ✖ I, III, V only
4. ✔ I, II, III, IV and V

Question Number : 16 Question Id : 630680395732 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which among the followings is NOT a metallic factor affecting chemical corrosion?

Options :

1. ✖ Electrode potential
2. ✖ Hydrogen over voltage of the surface
3. ✖ Presence of different phases in the surface of the metal
4. ✔ Rate of flow of oxygen

Question Number : 17 Question Id : 630680395733 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (application of metallic coating)	List-II (Use/ Application)
(A) Electroplating	(I) It is used for coating metals with films of metals having low melting point
(B) Hot dipping	(II) metals can be deposited as surface layer by allowing their vapours to strike metallic surfaces
(C) Vaporising	(III) Noble metals such as Au, Ag, W, Pt & base metals are used

Choose the correct answer from the options given below:

Options :

- ✗ A-(I), B- (II), C-(III)
- ✗ A-(I), B-(III), C-(II)
- ✓ A-(III), B-(I), C-(II)
- ✗ A-(III), B-(II), C-(I)

Question Number : 18 Question Id : 630680395734 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Statement-I: Thermosetting resins are formed by condensation polymerization.

Statement-II: Thermoplastics resins consist of long-chain linear polymers with negligible cross-links.

Options :

- ✓ Both the Statements are true.
- ✗ Both the Statements are false.
- ✗ Statement I is true but Statement II is false.
- ✗ Statement I is false but Statement II is true.

Question Number : 19 Question Id : 630680395735 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which among the following is NOT a thermoplastic polymer?

Options :

1. ✖ Polythene
2. ✖ PVC
3. ✔ Bakelite
4. ✖ Polystyrenes

**Question Number : 20 Question Id : 630680395736 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is NOT true for the rubber?

Options :

1. ✖ Natural rubber becomes soft at high temperature and brittle at low temperature.
2. ✖ Natural rubber is soluble in non-polar solvents and is non-resistant to attack by oxidizing agents.
3. ✖ The properties of the rubber are improved by heating it with sulphur and appropriate additive at a high temperature.
4. ✔ In the manufacture of tyre rubber, 25% of sulphur is used as a cross linking agent.

**Question Number : 21 Question Id : 630680395737 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

List-I (Polymer)	List-II (Use/Application)
(A) Polypropene	(I) As insulator, manufacturing of toys
(B) PVC	(II) Manufacture of raincoats, handbags
(C) Polystyrene	(III) Manufacture of ropes, fibers, pipes

Choose the correct answer from the options given below:

Options :

- ✗ A-(I), B- (II), C-(III)
- ✗ A-(I), B-(III), C-(II)
- ✗ A-(III), B-(I), C-(II)
- ✓ A-(III), B-(II), C-(I)

Question Number : 22 Question Id : 630680395738 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Statement-I: Calcium and Magnesium dissolved in water are two most common minerals that make water hard.

Statement-II: Hard water requires more soap and synthetic detergents for home laundry and washing.

Options :

- ✓ Both the Statements are true.
- ✗ Both the Statements are false.
- ✗ Statement-I is true but Statement-II is false.
- ✗ Statement-I is false but Statement-II is true.

Question Number : 23 Question Id : 630680395739 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The water with hardness 3.5 - 7.0 Grains per Gallon is:

Options :

1. ✖ Slightly hard
2. ✖ Hard
3. ✔ Moderately hard
4. ✖ Very hard

Question Number : 24 Question Id : 630680395740 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (Softening of water)	List-II (Involves the chemical)
(A) Lime- soda process	(I) Hydrated alumino silicates
(B) Zeolite or Permutit process	(II) Cation exchange resin
(C) Mixed bed deionization	(III) Ca(OH)_2

Choose the correct answer from the options given below:

Options :

1. ✖ A-(I), B- (II), C-(III)
2. ✖ A-(I), B-(III), C-(II)
3. ✔ A-(III), B-(I), C-(II)
4. ✖ A-(III), B-(II), C-(I)

Question Number : 25 Question Id : 630680395741 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The advantages of Reverse Osmosis are:

- I. It significantly reduces salt, most other inorganic material present in the water and some organic compounds.

II. It usually removes microscopic parasites.

III. It also removes pesticides, solvents and other volatile organic chemicals.

The correct statements from the above are:

Options :

1. ✖ I, II and III

2. ✔ I and II

3. ✖ II and III

4. ✖ I and III

Question Number : 26 Question Id : 630680395742 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The advantages of natural gas are:

I. It burns easily without any smoke. Hence it is very useful as a domestic fuel.

II. It has a very high calorific value (55 kJ/g) which is good for heating.

III. It causes no pollution or produces no poisonous gases.

IV. It can be supplied directly through pipes from the oil wells. This eliminates the need for storage and transport of the gases in cylinders.

The correct statements from the above are:

Options :

1. ✖ I, II and III only

2. ✖ I, III and IV only

3. ✔ I, II, III and IV

4. ✖ I and IV only

Question Number : 27 Question Id : 630680395743 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (Fuel)	List-II (Properties)
(A) Solid fuels	(I) Do not produce ash but sometimes produce smoke
(B) Liquid fuels	(II) They are least inflammable
(C) Gaseous fuels	(III) They are costly

Choose the correct answer from the options given below:

Options :

- ✓ A-(II), B- (I), C-(III)
- ✗ A-(I), B-(III), C-(II)
- ✗ A-(III), B-(I), C-(II)
- ✗ A-(III), B-(II), C-(I)

Question Number : 28 Question Id : 630680395744 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List-I (Name of Drug)	List-II (Use)
(A) Norethindrone	(I) Antibiotics
(B) Terpineol	(II) Antiseptics and disinfectants
(C) Ofloxacin	(III) Antifertility drug

Choose the correct answer from the options given below:

Options :

- ✗ A-(I), B- (II), C-(III)
- ✗ A-(I), B-(III), C-(II)
- ✗ A-(III), B-(I), C-(II)
- ✓ A-(III), B-(II), C-(I)

Question Number : 29 Question Id : 630680395745 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Statement-A: The air around the city of Agra, where the Taj Mahal is located, contains fairly high levels of sulphur and nitrogen oxides.

Statement-R: The government of India announced an action plan in early 1995 to prevent the disfiguring of the historical monument.

Options :

1. ✖ Both the statements are true and Statement R is correct explanation of Statement A
2. ✔ Both the statements are true but Statement R is NOT a correct explanation of Statement A
3. ✖ Statement A is true but Statement R is False.
4. ✖ Statement A is False but Statement R is True.

Question Number : 30 Question Id : 630680395746 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Chlorinated Hydrocarbons effects ecosystem by:

Options :

1. ✖ Bioremediation
2. ✔ Bioaccumulation
3. ✖ Biofiltration
4. ✖ Bioconversion

Question Number : 31 Question Id : 630680395747 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What is the correct order of 3R principle?

Options :

- 1.

✖ Reuse, Recycle, Reduce

2. ✔ Reduce, Reuse, Recycle

3. ✖ Recycle, Reduce, Reuse

4. ✖ Recycle, Reuse, Reduce

Question Number : 32 Question Id : 630680395748 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Protection of the ozone layer is related with:

Options :

1. ✖ Tokyo convention

2. ✔ Vienna Convention

3. ✖ Paris Convention

4. ✖ New York Convention

Question Number : 33 Question Id : 630680395749 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is an integral component of CPU?

Options :

1. ✔ Registers

2. ✖ Main memory

3. ✖ Secondary memory

4. ✖ Tertiary memory

Question Number : 34 Question Id : 630680395750 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following statements about input-output devices is True?

- (i) MICR can be used as an input as well as an output device.
- (ii) Both the scanner and the plotter are input devices.

Options :

- 1. ✖ (i) only
- 2. ✖ (ii) only
- 3. ✖ Both (i) and (ii)
- 4. ✔ Neither (i) nor (ii)

Question Number : 35 Question Id : 630680395751 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following category of virus saves itself in the memory of the computer and then infects other files and programs when its originating program is no longer working?

Options :

- 1. ✖ Boot sector virus
- 2. ✔ Resident virus
- 3. ✖ Polymorphic virus
- 4. ✖ Macro virus

Question Number : 36 Question Id : 630680395752 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is/are valid characteristics of an algorithm?

- (i) Definiteness
- (ii) Effectiveness

Options :

1. ✖ (i) only
2. ✖ (ii) only
3. ✔ Both (i) and (ii)
4. ✖ Neither (i) nor (ii)

Question Number : 37 Question Id : 630680395753 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following statements about network topologies is/are False?

- (i) Hubs or Switches are mainly used as connection devices in a physical star topology.
- (ii) In ring topology, the data flows in one direction, i.e., it is unidirectional.

Options :

1. ✖ (i) only
2. ✖ (ii) only
3. ✖ Both (i) and (ii)
4. ✔ Neither (i) nor (ii)

Question Number : 38 Question Id : 630680395754 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following network categories has smallest coverage area?

Options :

1. ✖ MAN
2. ✔ PAN
3. ✖ WAN

4. ✖ LAN

Question Number : 39 Question Id : 630680395755 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following network devices operates at the data link layer and used for interconnecting two LANs working on the same protocol?

Options :

1. ✖ Repeater

2. ✖ Hub

3. ✖ Switch

4. ✔ Bridge

Question Number : 40 Question Id : 630680395756 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The operating systems like iOS and Android are the examples of which of the following categories of operating system?

Options :

1. ✖ Real-time OS

2. ✔ Mobile OS

3. ✖ Time-sharing OS

4. ✖ Networked OS

Question Number : 41 Question Id : 630680395757 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following operating systems was released in 2015?

Options :

1. ✖ Windows Vista
2. ✖ Windows 7
3. ✖ Windows 8
4. ✔ Windows 10

Question Number : 42 Question Id : 630680395758 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is not a valid IPv4 address?

Options :

1. ✔ 10.100.200.300
2. ✖ 10.20.30.40
3. ✖ 200.100.50.10
4. ✖ 20.255.10.20

Question Number : 43 Question Id : 630680395759 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Database access language is one of the components of database systems. What is the full form of DML with respect to database access language?

Options :

1. ✖ Direct Manipulation Language
2. ✔ Data Manipulation Language
3. ✖ Data Mining Language

4. ✖ Data Morphing Language

Question Number : 44 Question Id : 630680395760 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Which of the following is/are benefits of e-commerce?

(i) Flexibility and Scalability

(ii) 24×7 Working

(iii) Cost-Effectiveness

Options :

1. ✖ (i) and (ii) only

2. ✖ (i) and (iii) only

3. ✖ (ii) and (iii) only

4. ✔ (i), (ii) and (iii)

Question Number : 45 Question Id : 630680395761 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Who is considered as a Father of Scientific Management?

Options :

1. ✖ W. A. Shewhart

2. ✔ Fredrick Winslow Taylor

3. ✖ Frank B. Gilbreth

4. ✖ Ralph M. Barnes

Question Number : 46 Question Id : 630680395762 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

PERT stands for _____.

Options :

1. ✖ Program examination review technique
2. ✖ Program exploration review technique
3. ✔ Program evaluation review technique
4. ✖ Program error review technique

**Question Number : 47 Question Id : 630680395763 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The activity related to supplier evaluation, selection, and purchasing the materials is known as _____.

Options :

1. ✖ advertising
2. ✔ outsourcing
3. ✔ sourcing
4. ✔ contracting

Note: For this question, ambiguity is found in question/answer. Candidate will get full marks for this question if any of the correct options are chosen.

**Question Number : 48 Question Id : 630680395764 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

_____ process provides the organization with adequate, competent, and qualified personnel at all levels in the enterprise.

Options :

- 1.

✖ Planning

2. ✔ Staffing

3. ✖ Organizing

4. ✖ Controlling

Question Number : 49 Question Id : 630680395765 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Three effective leadership includes the theories of _____, _____, _____ and _____.

Note: For this question, discrepancy is found in question/answer. So, this question is ignored for all candidates.

Options :

1. pacesetting, ethical, affiliative, behavioral
2. transformational, transactional, servant, full-range
3. democratic, autocratic, bureaucratic, charismatic, behavioral
4. trait, contingency, behavioral, full-range theories

Question Number : 50 Question Id : 630680395766 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List out the correct benefits of bin card which are used for keeping records?

- i) It gives up-to-date record of receipt, sits and closing balances of items of stores.
- ii) It is an account record which provides information about receipt, issue and balances both in quantity and value.
- iii) It is helpful in placing requisitions for replenishment as when necessary.

iv) It is maintained centrally in cost office from where consolidated information may be made available.

Options :

1. ✖ ii, iv

2. ✔ i, iii

3. ✖ i, ii

4. ✖ iii, iv

Question Number : 51 Question Id : 630680395767 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

_____ a map of the interior of the store that displays the various sections, shelves, or fixtures, and the ideal route for customer flow, all about the space available.

Options :

1. ✔ Store layout

2. ✖ Layout Planning

3. ✖ Decompression zone

4. ✖ Customer flow

Question Number : 52 Question Id : 630680395768 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Who will introduce the concept of “zero defects”?

Options :

1. ✖ Joseph M. Juran

2. ✔ Philip B. Crosby

3.

✖ Genichi Taguchi

4. ✖ Kaoru Ishikawa

Question Number : 53 Question Id : 630680395769 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

_____ may be defined as any condition produced by industries that may cause injury or death to personnel or loss of product or property.

Options :

1. ✔ Industrial Hazard

2. ✖ Hazard

3. ✖ Industrial Safety

4. ✖ Safety

Question Number : 54 Question Id : 630680395770 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List out the different approaches tools available for the cost analysis of accidents in the industries as follows:

- i) Insurance based approach
- ii) Medical cost based approach
- iii) Labour capacity based approach
- iv) Cost of damage based approach

Options :

1. ✖ ii, iv

2. ✔ i, iii

3. ✖ i, ii

4. ✖ ii, iii

Question Number : 55 Question Id : 630680395771 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

An _____ is an individual who creates a new business, bearing most of the risks and enjoying most of the rewards.

Options :

1. ✖ planning

2. ✖ supervisory

3. ✖ organizing

4. ✔ entrepreneur

Question Number : 56 Question Id : 630680395772 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

List out the correct overall goals of total quality management as:

I) Encouraging input from all employees

II) Preventive maintenance

III) Reduced training time and costs

IV) Maintaining equipment for increasing the life

Options :

1. ✔ I, IV

2. ✖ II, III

3. ✖ III, IV

4. ✖ II, IV

**Question Number : 57 Question Id : 630680395773 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

“Self-Employment Schemes” for the handicapped person who want to set up their business run by the Central Government and provides the financial benefits of _____.

Options :

1. ✖ a maximum limit of ₹10 Lakhs
2. ✖ a maximum limit of ₹30 Lakhs
3. ✔ a maximum limit of ₹50 Lakhs
4. ✖ a maximum limit of ₹70 Lakhs

**Question Number : 58 Question Id : 630680395774 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

List out the main objectives of management information system (MIS) as:

- i) defining quality standards for the deliverables.
- ii) provide a system of people, computers, procedures, interactive query facilities, documents for collecting, sorting, retrieving and transmitting information to the users.
- iii) making process by furnishing information in the proper time frame.
- iv) gaining the customers agreement with the targets set.

Options :

1. ✖ i, iv
2. ✖ iii, iv
3. ✔ ii, iii
4. ✖ i, ii

Question Number : 59 Question Id : 630680395775 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

_____ was first published in 1987 by the International Organization for Standardization (ISO).

Options :

1. ✖ ISO 9001

2. ✖ ISO 9004

3. ✖ ISO 19011

4. ✔ ISO 9000

Question Number : 60 Question Id : 630680395776 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In which year Micro, Small, and Medium Enterprises was formed in India?

Options :

1. ✖ 1959

2. ✔ 1961

3. ✖ 1963

4. ✖ 1965

Question Number : 61 Question Id : 630680395777 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

A driving force behind the rapid development of _____ technology has been the rise of pervasive commerce, sometimes dubbed the quiet revolution.

Options :

1. ✖ Information and Communication Technology

2. ✖ Electronic Article Surveillance
3. ✖ Surface-Mounting technology
4. ✔ Radio Frequency Identification

**Question Number : 62 Question Id : 630680395778 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

Identify the correct benefits of preventive maintenance from below:

- i) Less equipment downtime
- ii) Shorter asset life
- iii) Longer asset life
- iv) More failure equipment

Options :

1. ✖ i, ii
2. ✔ i, iii
3. ✖ ii, iii
4. ✖ iii, iv

**Question Number : 63 Question Id : 630680395779 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

PDSA cycles for improvement in six sigma stands for _____.

Options :

1. ✖ Planning-Do-Strategi-Act
2. ✖ Plan-Document-Study-Act
3. ✖ Planning-Development-System-Act

4. ✓ Plan-Do-Study-Act

Question Number : 64 Question Id : 630680395780 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

There are seven basic tools of quality control are histogram, Pareto diagram, check sheet, flowchart, scatter diagram, _____ and _____. Find out the missing two tools for quality control method.

Options :

1. ✗ Total quality management diagram, Pie charts,
2. ✓ Cause and effect diagram, Control charts
3. ✗ Matrix diagram, Industrial management flow charts
4. ✗ Total quality management diagram, Gantt charts

Question Number : 65 Question Id : 630680395781 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In an IT company follows a piece rate system of incentive with guaranteed minimum wage for a particular project. On the basis of hourly work rate is ₹ 15 only. The piece rate for a particular type of job is ₹ 10. An average worker can make only 10 jobs a day (i.e. 8 hours a day). Determine an average worker's daily earning.

Options :

1. ✓ ₹ 220/day
2. ✗ ₹ 320/day
3. ✗ ₹ 120/day
4. ✗ ₹ 420/day

Question Number : 66 Question Id : 630680395782 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

An operator was kept under observation for 30 days. He was found working on 800 occasions and abstaining including idle time was 200. He produced 350 jobs during these days. The observation per day was for 6 hours only and the total number of observations was 1000. Take 110 as performance rating for operator and 30 per cent as allowance. Determine the standard time for an operator.

Options :

1. ✖ 15.30 minutes

2. ✖ 25.30 minutes

3. ✔ 35.30 minutes

4. ✖ 45.30 minutes

Question Number : 67 Question Id : 630680395783 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Find out the correct disadvantages of sampling plan:

i) Convenience

ii) Sampling error

iii) Feasibility

iv) Time constraints

Options :

1. ✔ ii, iv

2. ✖ iii, iv

3. ✖ i, ii

4. ✖ ii, iii

**Question Number : 68 Question Id : 630680395784 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

_____ is a statistical technique used to inspect the quality of a batch of products.

Options :

1. ✖ Time-consuming
2. ✖ Statistical sampling
3. ✖ Quality control technique
4. ✔ Acceptance sampling

**Question Number : 69 Question Id : 630680395785 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

If 500 literature books are inspected and 20 books are found defective. Determine the fraction defective of the literature book is given as :

Options :

1. ✖ 0.02
2. ✔ 0.04
3. ✖ 0.06
4. ✖ 0.08

**Question Number : 70 Question Id : 630680395786 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

Find out the correct types of on-site quality inspection services are:

- i) Initial Production Check

- ii) Generates waste
- iii) During Production Inspection
- iv) Acceptable quality limit

Options :

1. ✓ i, iii

2. ✗ ii, iv

3. ✗ i, ii

4. ✗ ii, iii

Question Number : 71 Question Id : 630680395787 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The control chart for process mean is the _____.

Options :

1. ✓ $\bar{X}$ -chart

2. ✗ R-chart

3. ✗ S-chart

4. ✗ p-chart

Question Number : 72 Question Id : 630680395788 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Who is introduce the “Skill and Effort Rating” system in work measurement techniques?

Options :

1. ✖ M. E. Mundel

2. ✖ Sanders and McCormick

3. ✔ Charles E. Bedaux

4. ✖ Bedaux plan

Question Number : 73 Question Id : 630680395789 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Suppose X can copy 100 pages in 25 hours while X and Y together can copy 150 pages in 30 hours. Then Y can copy 23 pages in how many hours?

Options :

1. ✖ 13

2. ✖ 17

3. ✔ 23

4. ✖ 29

Question Number : 74 Question Id : 630680395790 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

A train crosses 110 m long platform in 12 s at the speed of 54 km/hr. What is the length of the train (in m)?

Options :

1. ✖ 60

2. ✔ 70

3. ✖ 80

4. ✖ 90

**Question Number : 75 Question Id : 630680395791 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

A man walks from his house at an average speed of 6 km/hr and reaches his office 5 minutes late. If he walks at an average speed of 7 km/hr, he could reach 3 minutes early. What is distance of the office from his house (in km)?

Options :

- 1. ✓ 5.6
- 2. ✗ 6.5
- 3. ✗ 8.5
- 4. ✗ 10.6

**Question Number : 76 Question Id : 630680395792 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

Suppose the price of an item was increased by 25%. Then the price was decreased by 20% and again increased by 10%. What is the resultant increase in the price of the item?

Options :

- 1. ✗ 6%
- 2. ✗ 8%
- 3. ✓ 10%
- 4. ✗ 12%

**Question Number : 77 Question Id : 630680395793 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

Suppose zinc and copper are in the ratio of 5 : 4 in 225 gm of an alloy. How much grams of copper

should be added to make the ratio as 4 : 5?

Options :

1. ✓ $56\frac{1}{4}$

2. ✗ $60\frac{1}{4}$

3. ✗ 56

4. ✗ 60

Question Number : 78 Question Id : 630680395794 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

On selling an article for Rs. 360, there is a loss of 10%. What is the cost price of that article (in Rs.)?

Options :

1. ✗ 300

2. ✓ 400

3. ✗ 500

4. ✗ 550

Question Number : 79 Question Id : 630680395795 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In how many times (in years) will the simple interest be $\frac{2}{5}$ of the principal at 8% per annum?

Options :

1. ✖ 3

2. ✖ 3.5

3. ✔ 5

4. ✖ 5.5

Question Number : 80 Question Id : 630680395796 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Suppose the incomes of two persons A and B are in the ratio 3 : 2 and their expenditures are in the ratio 5:3. If each saves Rs. 1,000, then match column I with column II in the following table:

Column I	Column II
P: A's income	1. Rs. 3000
Q: B's income	2. Rs. 6000
R: A's expenditure	3. Rs. 4000
S: B's expenditure	4. Rs. 5000

Options :

1. ✖ P-1; Q-2; R-3; S-4

2. ✖ P-3; Q-2; R-1; S-4

3. ✔ P-2; Q-3; R-4; S-1

4. ✖ P-4; Q-3; R-2; S-1

Question Number : 81 Question Id : 630680395797 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Based on the Euclid's division algorithm, consider the following assertion and reason statements:

Assertion (A): HCF of 81 and 675 using the Euclid's division algorithm is 27.

Reason (R): $675 = 81 \times 8 + 27$

$$81 = 27 \times 3 + 0$$

Then which of the options is correct according to the Euclid's division algorithm?

Options :

1. ✓ both A and R are true, and R is the correct explanation of A
2. ✗ both A and R are true, but R is not the correct explanation of A
3. ✗ A is true but R is false
4. ✗ both A and R are false

Question Number : 82 Question Id : 630680395798 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

If $a = b^{-\frac{2}{3}}$ and $b = c^{-3}$, then the value of c in terms of a is:

Options :

1. ✓ $\sqrt{a}$
2. ✗ $\sqrt[3]{a}$
3. ✗ a^2
4. ✗ a^3

Question Number : 83 Question Id : 630680395799 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What is the value of $\sqrt{0.01} + \sqrt{2.89} + \sqrt{1.21} + \sqrt{0.0009}$?

Options :

1. ✖ 2.53

2. ✖ 2.73

3. ✖ 2.91

4. ✔ 2.93

Question Number : 84 Question Id : 630680395800 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

How many positive factors of 200 are there?

Options :

1. ✖ 6

2. ✖ 9

3. ✔ 12

4. ✖ 15

Question Number : 85 Question Id : 630680395801 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Let the HCF and LCM of two numbers be 11 and 363, respectively. If one of the numbers lies between 111 and 129, then what is that number?

Options :

1. ✖ 113

2. ✖ 117

3.

✖ 119

4. ✔ 121

Question Number : 86 Question Id : 630680395802 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What is the largest length (in cm) of the tape to be used to measure the room sides whose distances are 6 m 48 cm, 11 m 52 cm and 16 m 20 cm?

Options :

1. ✖ 72

2. ✔ 36

3. ✖ 32

4. ✖ 45

Question Number : 87 Question Id : 630680395803 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What number should be added to 231228 in order to make it exactly divisible by 11?

Options :

1. ✖ 1

2. ✖ 2

3. ✔ 3

4. ✖ 4

Question Number : 88 Question Id : 630680395804 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The value of $[\log_{10}(50\log_{10}100)]^3$ is:

Options :

1. ✖ 16

2. ✔ 8

3. ✖ 4

4. ✖ 1

Question Number : 89 Question Id : 630680395805 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

$9^{3.5}:2^2 \times 3^5$ is the same as:

Options :

1. ✖ 3:2

2. ✖ 9:2

3. ✖ 3:4

4. ✔ 9:4

Question Number : 90 Question Id : 630680395806 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

When a polynomial $f(x) = ax^3 - bx + r$ is divided by $g(x)$, it leaves quotient as bx , and remainder as r . Then what is the value of $g(x)$?

Options :

1. ✔ $ax^2 - 1$

2. ✖ $x^2 - 1$

3. ✖ $ax^2 + 1$

4. ✖ $-ax^2 + 1$

Question Number : 91 Question Id : 630680395807 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Based on the linear equations in two variables, consider the following assertion and reason statements. Then which of the options is correct?

Assertion (A): The equation of degree one in two variables is called a linear equation.

Reason (R): The graph of an equation of degree one in two variables is a circle.

Options :

- 1. ✖ both A and R are true, and R is the correct explanation of A
- 2. ✖ both A and R are true, but R is not the correct explanation of A
- 3. ✔ A is true but R is false
- 4. ✖ both A and R are false

Question Number : 92 Question Id : 630680395808 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

If two roots of the polynomial $x^3 - 2x^2 - 13x - 10$ are -2 and 5 , then the third root is:

Options :

1. ✖ -3

2. ✖ -2

3. ✖ 1

4. ✔ -1

**Question Number : 93 Question Id : 630680395809 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

What is the sum of all positive integers lying between 200 and 400 that are multiples of 7?

Options :

1. ✖ 8703

2. ✔ 8729

3. ✖ 8297

4. ✖ 8792

**Question Number : 94 Question Id : 630680395810 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The values of k for which the quadratic equation $(3k + 1)x^2 + 2(k + 1)x + 1 = 0$ has equal roots are:

Options :

1. ✔ 0, 1

2. ✖ 0, 2

3. ✖ 0, -1

4. ✖ $1, -2$

Question Number : 95 Question Id : 630680395811 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Suppose the marks scored by a student in a mathematics test are 17, 12, 7, 2, 15, 8, 14, 18, 11, 14, 18, 10, 4, 7, 8, 5. Then what is the median of this data?

Options :

1. ✖ 8

2. ✖ 10

3. ✔ 10.5

4. ✖ 18

Question Number : 96 Question Id : 630680395812 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Based on the circle, consider the following assertion and reason statements. Then which of the options is correct?

Assertion (A): Through three collinear points, a circle can be drawn.

Reason (R): A circle can pass through only two collinear points as well as through three collinear points.

Options :

1. ✖ both A and R are true, but R is not the correct explanation of A

2. ✖ both A and R are true, and R is the correct explanation of A

3. ✖ A is true but R is false

4. ✔ both A and R are false

Question Number : 97 Question Id : 630680395813 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The common tangent of the two touching circles $2x^2 + 2y^2 + 6x - 2y - 7 = 0$ and $2x^2 + 2y^2 - 4x + 7y + 9 = 0$ is:

Options :

1. ✓ $10x - 9y - 16 = 0$

2. ✗ $10x + 9y + 16 = 0$

3. ✗ $10x - 9y + 16 = 0$

4. ✗ $10x + 9y - 16 = 0$

Question Number : 98 Question Id : 630680395814 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What will be the coordinates of B, if the point $C\left(\frac{16}{5}, \frac{13}{5}\right)$, divides the line segment joining A (3,6) and B (a,b) in the ratio 2 : 3?

Options :

1. ✗ $a = -\frac{7}{2}, b = \frac{5}{2}$

2. ✓ $a = \frac{7}{2}, b = -\frac{5}{2}$

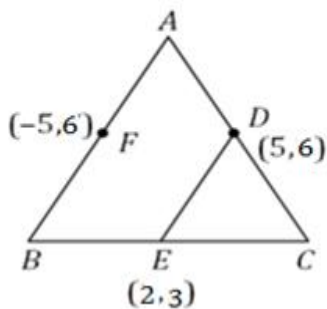
3. ✗ $a = -\frac{7}{2}, b = -\frac{5}{2}$

4. ✖ $a = \frac{7}{2}, b = \frac{5}{2}$

Question Number : 99 Question Id : 630680395815 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

If AB is parallel to DE as shown in the following figure, then what is the equation of the line AB?



Options :

1. ✖ $x - 2y - 11 = 0$

2. ✖ $2x + y + 1 = 0$

3. ✔ $x - y + 11 = 0$

4. ✖ $x + y - 1 = 0$

Question Number : 100 Question Id : 630680395816 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Select the option that is true regarding the following two statements labelled Assertion and Reason.

Assertion: In ΔABC , $BC = AB$ and $\angle B = 80^\circ$. Then, $\angle A = 50^\circ$

Reason: In a triangle, angles opposite to two equal sides are equal

Options :

1. ✓ Assertion and reason are true and reason is the correct explanation of assertion.
2. ✗ Both assertion and reason are true but reason is not the correct explanation of assertion.
3. ✗ Assertion is true but reason is false.
4. ✗ Assertion is false but reason is true.

Question Number : 101 Question Id : 630680395817 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Select the option that is true regarding the following two statements labelled Assertion and Reason.

Assertion: If the diagonals of a parallelogram ABCD are equal, then $\angle ABC = 90^\circ$.

Reason: If the diagonals of a parallelogram are equal, it becomes a rectangle.

Options :

1. ✗ Assertion is false but reason is true
2. ✗ Both assertion and reason are true but reason is not the correct explanation of assertion
3. ✗ Assertion is true but reason is false
4. ✓ Assertion and reason are true and reason is the correct explanation of assertion

Question Number : 102 Question Id : 630680395818 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

If $\sin A = \frac{1}{2}$ and $\sin B = \frac{\sqrt{3}}{2}$, then $A + B = ?$

Options :

1. ✗ 30°
2. ✗ 45°
3. ✗ 60°

4. ✓ 90°

**Question Number : 103 Question Id : 630680395819 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The angle of elevation of a tower from a distance 75 m from its foot is 60°. Height of the tower (in m) is:

Options :

1. ✗ $50\sqrt{3}$

2. ✓ $75\sqrt{3}$

3. ✗ $\frac{75}{\sqrt{3}}$

4. ✗ $\frac{150}{\sqrt{3}}$

**Question Number : 104 Question Id : 630680395820 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The value of the expression:

$$\left\{ \frac{\sec^2 63^\circ + \sec^2 27^\circ}{\operatorname{cosec}^2 63^\circ + \operatorname{cosec}^2 27^\circ} + \frac{1}{2} (\sec^2 25^\circ - \tan 25^\circ \cdot \cot 65^\circ) \right\}$$

Options :

1. ✓ $\frac{3}{2}$

2. ✖ 0

3. ✖ 1

4. ✖ $\frac{1}{2}$

Question Number : 105 Question Id : 630680395821 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33

If $\cos \alpha = \frac{2}{3}$, then match the pair given in column A with correct value in column B:

Column A	Column B
A) $\tan \alpha$	1) $\frac{2}{\sqrt{5}}$
B) $\operatorname{cosec} \alpha$	2) $\frac{\sqrt{5}}{2}$
C) $\cot \alpha$	3) $\frac{3}{\sqrt{5}}$

Options :

1. ✖ A - 1, B - 2, C - 3

2. ✖ A - 2, B - 1, C - 3

3. ✔ A - 2, B - 3, C - 1

4. ✖ A - 3, B - 1, C - 2

Question Number : 106 Question Id : 630680395822 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33

Find the area (in sq. cm) of a triangle two sides of which are 15 cm and 12 cm and the perimeter is 40 cm.

Options :

1. ✖ $10\sqrt{7}$
2. ✖ $10\sqrt{14}$
3. ✖ $20\sqrt{7}$
4. ✔ $20\sqrt{14}$

Question Number : 107 Question Id : 630680395823 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Select the option that is true regarding the following two statements labelled Assertion and Reason.

Assertion (A): Area of the square inscribed in a circle of radius r is $2r^2$ sq . Unit.

Reason (R): Area of the major segment of a circle = area of the circle - Area of minor segment.

Options :

1. ✖ Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
2. ✔ Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
3. ✖ Assertion (A) is true but Reason (R) is false
4. ✖ Assertion (A) is false but Reason (R) is true

Question Number : 108 Question Id : 630680395824 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

What is the height of a cone (in inches) whose radius is 7 inches and curved surface area is 550 in^2 ?

Options :

1. ✖ 25

2. ✔ 24

3. ✖ 23

4. ✖ 22

Question Number : 109 Question Id : 630680395825 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

State true or false.

a.) The sum of areas of all 6 side faces of a cube is its lateral area.

b.) The formula to find the lateral area of a cube is, $LSA = 4x^2$ (where x is side of the cube).

Options :

1. ✖ a-TRUE, b-TRUE

2. ✖ a-TRUE, b-FALSE

3. ✔ a-FALSE, b-TRUE

4. ✖ a-FALSE, b-FALSE

Question Number : 110 Question Id : 630680395826 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The volume of a sphere has grown 8 times. Then how many times has the surface area grown in the meanwhile?

Options :

1. ✖ 2

2. ✔ 4

3. ✖ 8

4. ✖ 16

Question Number : 111 Question Id : 630680395827 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

A breakfast cereal producer wants to produce a cereal box with a volume of 24000 cubic cm. if dimensions of cuboid are in ratio 2 : 3 : 4, then what are the dimensions (in cm)?

Options :

1. ✔ 20, 30, 40

2. ✖ 20, 40, 60

3. ✖ 10, 20, 30

4. ✖ 15, 30, 45

Question Number : 112 Question Id : 630680395828 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In an inelastic collision, the physical quantity that remains constant is _____.

Options :

1. ✖ Kinetic Energy

2. ✔ Momentum

3. ✖ Potential Energy

4. ✖ Force

Question Number : 113 Question Id : 630680395829 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

A person moving through a distance of 5m is carrying of 5 N. The force carried by the person is perpendicular to the displacement made by him. The work done by him is _____.

Options :

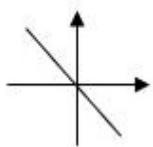
1. ✖ Positive
2. ✖ Negative
3. ✔ Zero
4. ✖ Unity

Question Number : 114 Question Id : 630680395830 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The following graph plotted between force and displacement represents _____.



Options :

1. ✖ Linear Motion
2. ✔ Simple Harmonic Motion
3. ✖ Circular Motion
4. ✖ Curvilinear Motion

Question Number : 115 Question Id : 630680395831 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Given below are two statements. Identify the CORRECT statement(s).

Statement A: The periodic increase and decrease of loudness of sound when two sounds of nearly equal frequencies interfere with each other is called the Doppler Effect.

Statement B: The apparent change in frequency of sound due to relative motion between source and observer produces beats.

Options :

1. ✖ Statement A and Statement B are correct
2. ✖ Statement A is correct and Statement B is incorrect
3. ✖ Statement B is correct and Statement A is incorrect
4. ✔ Both the statements A and B are incorrect

Question Number : 116 Question Id : 630680395832 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The susceptibility of a superconductor is _____.

Options :

1. ✖ Positive and small
2. ✖ Positive and high
3. ✖ Zero
4. ✔ Negative

Question Number : 117 Question Id : 630680395833 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

As the size of the particle is reduced to nano level, then _____.

Options :

1. ✔ Surface to volume ratio of atoms increases

2. ✖ Surface to volume ratio of atoms is zero
3. ✖ Surface to volume ratio of atoms does not change
4. ✖ Surface to volume ratio of atoms decreases.

**Question Number : 118 Question Id : 630680395834 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

Frictional force is _____.

Options :

1. ✖ Conservative
2. ✖ always zero
3. ✔ Non-conservative
4. ✖ always constant in magnitude

**Question Number : 119 Question Id : 630680395835 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

In graded -index fibers, the refractive index of core _____.

Options :

1. ✖ Is uniform
2. ✖ Increases with radial distance from the axis of the fiber
3. ✖ Is constant
4. ✔ decreases with radial distance from the axis of the fiber

**Question Number : 120 Question Id : 630680395836 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The magnitude of three vectors is such that $P + Q = R$. Then the angle between P and Q is _____.

Options :

- 1. ✖ 0°
- 2. ✖ 45°
- 3. ✖ 60°
- 4. ✔ 90°

Question Number : 121 Question Id : 630680395837 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The S.I. unit of coefficient of viscosity is _____.

Options :

- 1. ✔ Nm^{-2}s
- 2. ✖ Nm^{-1}s
- 3. ✖ Nm^{-2}
- 4. ✖ Nms^{-1}

Question Number : 122 Question Id : 630680395838 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The body with higher Kinetic energy when a plastic ball and an iron ball are moving with a momentum of 5 Kgms^{-1} is _____. (Here mass of iron ball is greater than the mass of plastic ball)

Options :

- 1. ✔ Plastic ball
- 2. ✖ Iron ball
- 3.

✖ Kinetic energy of iron ball is equal to that of plastic ball

4. ✖ Kinetic energy of iron ball and plastic ball is zero

Question Number : 123 Question Id : 630680395839 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The force acting between two charges q_1 and q_2 separated by a distance is F . What is the direction of force acting between the charges?

Options :

1. ✖ It acts at an angle of 90° to the charges

2. ✖ It acts along an angle of 45° to the charges

3. ✖ It acts at an angle of 60° to the two charges

4. ✔ It acts along the line joining the two charges

Question Number : 124 Question Id : 630680395840 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Given below are two statements. Identify the CORRECT statement(s).

Statement A: The inability of a body to change its position from rest or of uniform motion unless acted by an external force is called Inertia.

Statement B: Inertia is independent of mass of the body.

Options :

1. ✖ Statement A and Statement B are correct

2. ✔ Statement A is correct and Statement B is incorrect

3. ✖ Statement A is incorrect and Statement B is correct

4. ✖ Both the statements A and B are incorrect

**Question Number : 125 Question Id : 630680395841 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

The angle between the vectors whose scalar product is zero is _____.

Options :

- 1. ✖ 0°
- 2. ✖ 45°
- 3. ✖ 60°
- 4. ✔ 90°

**Question Number : 126 Question Id : 630680395842 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

The velocity of sound is maximum in _____.

Options :

- 1. ✔ Iron
- 2. ✖ Water
- 3. ✖ Oxygen
- 4. ✖ Vacuum

**Question Number : 127 Question Id : 630680395843 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1 Wrong Marks : 0.33**

Given below are two statements. Identify the CORRECT statement(s).

Statement A: Unit of Force in S.I. Unit is Kgms^{-1}

Statement B: Unit of Power in S.I. unit is Kgms^{-3}

Options :

1. ✖ Statement A and Statement B are correct
2. ✖ Statement A is correct and Statement B is incorrect
3. ✔ Both the statements A and B are incorrect
4. ✖ Statement A is incorrect and Statement B is correct

Question Number : 128 Question Id : 630680395844 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Match the following:

Column-I		Column-II	
a)	Zeroeth law of Thermodynamics	1)	Transfer of heat from cold body to hot body takes place only with the aid of external agency
b)	First law of Thermodynamics	2)	Concept of temperature
c)	Kelvin's Statement of Second law of Thermodynamics	3)	Heat and Work are interconvertible.
d)	Clasius Statement of Second law of Thermodynamics	4)	Impossible to derive work by reducing the temperature of source colder than the coldest of its surroundings

Options :

1. ✔ a - 2; b - 3; c - 4; d - 1
2. ✖ a - 3; b - 2; c - 4; d - 1
3. ✖ a - 2; b - 3; c - 1; d - 4
4. ✖ a - 3; b - 2; c - 1; d - 4

Question Number : 129 Question Id : 630680395845 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The S.I. unit of magnetic field is _____.

Options :

1. ✖ Ampere

2. ✔ Tesla

3. ✖ Henry

4. ✖ Oersted

Question Number : 130 Question Id : 630680395846 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The Young's modulus (Y) of a wire of initial length 'L' and area of cross-section 'A' undergoing an extension 'l', when a stretching force 'F' is applied in the direction of its length is given by $Y = \frac{FL}{Al}$.

Options :

1. ✔ $\frac{FL}{Al}$

2. ✖ $\frac{FA}{Li}$

3. ✖ $\frac{Fl}{AL}$

4. ✖ $\frac{FLI}{A}$

Question Number : 131 Question Id : 630680395847 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

One mole of a gas at a temperature of 27°C having a volume of 150 ml is compressed isothermally to 75 ml. The final temperature of the gas is _____.

Options :

1. ✖ 0 K

2. ✖ 150 K

3. ✔ 300 K

4. ✖ 600 K

Question Number : 132 Question Id : 630680395848 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Two particles of masses m_1 and m_2 are moving in concentric circles of radii r_1 and r_2 such that their periods are same. The ratio of their centripetal acceleration is _____.

Options :

1. ✖ r_1^2/r_2^2

2. ✖ r_2^2/r_1^2

3. ✔ r_1/r_2

4. ✖ r_2/r_1

Question Number : 133 Question Id : 630680395849 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

According to Poiseuille's equation, the viscosity, η is proportional to (Here 'r' is the radius of capillary bore)

Options :

1. ✖ r^2

2. ✓ r^4

3. ✗ $1/r^2$

4. ✗ $1/r^4$

**Question Number : 134 Question Id : 630680395850 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

A force of 50 N is applied on a perfectly elastic body of length 10 m along its length. When the force is removed, the length becomes _____.

Options :

1. ✗ 5 m

2. ✗ 5 cm

3. ✗ 10 cm

4. ✓ 10 m

**Question Number : 135 Question Id : 630680395851 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The linear elongation of a material of mass 5 Kg is elongated to 2 m. If the lateral contraction of the body is 0.5 m, the Poisson's ratio is _____.

Options :

1. ✗ 20 m

2. ✗ 1.25 m

3. ✗ 4 m

4. ✓ 0.25 m

**Question Number : 136 Question Id : 630680395852 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The horizontal range of a projectile is maximum, when the body is projected at an angle of _____.

Options :

1. ✓ 45°

2. ✗ 90°

3. ✗ 0°

4. ✗ 135°

**Question Number : 137 Question Id : 630680395853 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The dimensional formula of Impulse is _____.

Options :

1. ✗ MLT^{-2}

2. ✓ MLT^{-1}

3. ✗ $ML^{-1}T^{-1}$

4. ✗ $ML^{-2}T^{-2}$

Question Number : 138 Question Id : 630680395854 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In Photoelectric effect, photoelectrons are emitted with zero kinetic energy, when _____.

Here, ν is the applied frequency and ν_0 is the threshold frequency

Options :

1. ✓ $\nu = \nu_0$

2. ✗ $\nu > \nu_0$

3. ✗ $\nu < \nu_0$

4. ✗ $\nu = 0$

Question Number : 139 Question Id : 630680395855 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

When the body is at rest, its _____ is maximum.

Options :

1. ✗ Kinetic Energy

2. ✓ Potential Energy

3. ✗ Velocity

4. ✗ Acceleration

Question Number : 140 Question Id : 630680395856 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Ohm's law is valid only for _____.

Options :

1. ✓ Copper
2. ✗ Germanium
3. ✗ Mica
4. ✗ Arsenic

**Question Number : 141 Question Id : 630680395857 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The unit of stress is _____.

Options :

1. ✗ Newton
2. ✓ Pascal
3. ✗ Nm^2
4. ✗ Pascal/m^2

**Question Number : 142 Question Id : 630680395858 Is Question Mandatory : No Calculator :
None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0**

Correct Marks : 1 Wrong Marks : 0.33

The following statement is correct in case of Noise -

Options :

1. ✗ Vibrations of the particles of the medium are regular
2. ✗ The pitch and loudness are within the specified limits
3. ✓ There will be sudden change in amplitude
4. ✗ Pitch is within the specific limits

Question Number : 143 Question Id : 630680395859 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

In an optical fiber, the refractive index of core is n_1 and the refractive index of cladding is n_2 . The light gets transmitted through the fiber without any loss when _____.

Options :

1. ✖ $n_1/n_2 < 1$

2. ✖ $n_1/n_2 = 0$

3. ✔ $n_1/n_2 > 1$

4. ✖ $n_1 n_2 = 0$

Question Number : 144 Question Id : 630680395860 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

One mole gas at temperature 27°C is compressed isothermally from a volume of 20 litre to 10 litre. The final temperature of the gas is _____.

Options :

1. ✔ 300 K

2. ✖ 150 K

3. ✖ 600 K

4. ✖ 75 K

Question Number : 145 Question Id : 630680395861 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Two forces 15 N and 8 N are perpendicular to each other. The resultant of the two vectors is _____.

Options :

1. ✖ 23 N
2. ✖ 7 N
3. ✔ 17 N
4. ✖ 529 N

Question Number : 146 Question Id : 630680395862 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Match the following related to Simple Harmonic Motion.

Column-I		Column-II	
a)	Acceleration is maximum at	1)	$-kx$
b)	Velocity is maximum at	2)	equilibrium position
c)	Restoring forces in SHM	3)	$-\omega^2x$
d)	Acceleration of a particle executing SHM	4)	extreme position

Options :

1. ✔ a - 4; b - 2; c - 1; d - 3
2. ✖ a - 2; b - 4; c - 3; d - 1
3. ✖ a - 4; b - 2; c - 3; d - 1
4. ✖ a - 2; b - 4; c - 1; d - 3

Question Number : 147 Question Id : 630680395863 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Let H be the maximum height reached by the projectile. At the highest point the physical quantity that becomes zero is _____.

Options :

1. ✓ Vertical velocity
2. ✗ Angle of projection
3. ✗ Horizontal velocity
4. ✗ Vertical distance

Question Number : 148 Question Id : 630680395864 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

The specific resistance of the given wire of radius 'r' from metre bridge is proportional to _____.

Options :

1. ✗ R
2. ✗ $1/r$
3. ✓ r^2
4. ✗ $1/r^2$

Question Number : 149 Question Id : 630680395865 Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

If m_1 , m_2 are masses of two particles, separated by a distance 'd' then the force of attraction between the particles is 'F'. When both the masses of the particles are doubled and the distance becomes (d/2), then the new force of attraction between the bodies will be _____.

Options :

1. ✓ 16 F

2. ✗ 4 F

3. ✗ 2 F

4. ✗ F

Question Number : 150 Question Id : 630680395866 Is Question Mandatory : No Calculator :

None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1 Wrong Marks : 0.33

Vanderwaal's gases deviate from ideal gas behavior because _____

Options :

1. ✓ The molecules of the gas interact with each other by collisions

2. ✗ No interaction takes place between the molecules of a gas

3. ✗ Individual motion of molecules does not exist

4. ✗ Their kinetic energy becomes zero